

Quick Info

- Candidate: [Your Name]
- Lab: [Research Lab]
- Duration: 3–6 months
- Theme: Vibration analysis and control

1. Research Question

Can we detect and classify structural anomalies (e.g., looseness, cracks, imbalance) from vibration signatures using a physics-informed workflow combining simulation and signal processing?

2. Methodology

- Acquire or simulate vibration data (simple beam/rotor model).
- Extract features: FFT peaks, envelopes, time–frequency representations.
- Compare baseline vs faulty conditions; validate repeatability.
- Optional: train a lightweight classifier for fault identification.

3. Deliverables

- Simulation model + dataset.
- Feature extraction pipeline.
- Final report with recommendations for future work.